

SUPPORTING INFORMATION for:

Modeling of tumor growth with input from patient-specific metabolomic data

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1. Simulation of Vascularized Tumor Growth

1.1 Tumor Growth

The tumor growth component is based on ^{5, 12} and represents tumor tissue in three regions: a proliferating region where cells have sufficient oxygen and nutrients to proliferate, a hypoxic region where cells have sufficient oxygen and nutrients to survive but insufficient for proliferation, and a necrotic region where cells lack sufficient oxygen and nutrients for survival.

The non-dimensionalized tumor growth velocity follows Darcy's Law ⁵:

$$\mathbf{v}_c = -\mu \nabla P + \chi_E \nabla E \quad (1)$$

where μ is cell-mobility, which models the net effects of cell-cell and cell-matrix adhesion, P is oncotic pressure, χ_E is haptotaxis, and E is the density of the extracellular matrix (ECM). Refer to ⁵ for a more detailed description of E and χ_E . The overall tumor growth can be associated with the rate of volume change by assuming that the cell density within the proliferating region remains constant:

$$\nabla \cdot \mathbf{v}_c = \lambda_p \quad (2)$$

where λ_p is the non-dimensionalized net proliferation rate (described below).

1.2 Tumor Vascularization

The angiogenesis component is based on ⁸ to represent blood flow, vascular leakage and vascular network remodeling resulting from wall shear stress and mechanical stress imposed by the tumor tissue as it changes in time. Briefly, as the tumor grows within a vascularized environment, the tissue has access to oxygen and nutrients diffusing from the vasculature. The interstitial flow of oxygen and nutrients is influenced by tissue pressure and by distance from the nearest vessel. Neo-vessel sprouts arise from the host vasculature, whose epithelial cell tips migrate semi-stochastically towards a gradient of tumor angiogenic factors (TAF), which are produced by tumor tissue:

$$0 = \nabla \cdot (D_T \nabla T) + \bar{\lambda}_{production}^T (1 - T) \mathbf{1}_{\Omega_H} - \bar{\lambda}_{binding}^T \mathbf{1}_{vessel} - \bar{\lambda}_{ECM\ uptake}^T ET - \bar{\lambda}_{decay}^T T \quad (3)$$

The TAF T diffuse through the tissue with coefficient D_T , are produced in the tumor hypoxic region with a rate $\bar{\lambda}_{production}^T$, bind to the surrounding capillary vessels with a rate $\bar{\lambda}_{binding}^T$, are uptaken by the ECM (with density E) with a rate $\bar{\lambda}_{ECM\ uptake}^T$, and decay with a rate $\bar{\lambda}_{decay}^T$. Refer to ^{5, 12} for a more detailed description of the angiogenesis model.

1.3 Transport of Oxygen and Nutrients

Oxygen and nutrients σ are transported to the tumor from the location of extravasation from the vasculature. The extravasation rates $\lambda_{ev}^\sigma = \lambda_{neo}^\sigma$ and $\lambda_{ev}^\sigma = \lambda_{pre}^\sigma$ represent the rate that σ are supplied from the neo- and pre-existing vasculature, respectively. These substances, which diffuse with a coefficient D_σ , are respectively taken up by host tissue cells, and proliferating and hypoxic tumor cells, with rates $\lambda_{host}^\sigma, \lambda_{prolif}^\sigma, \lambda_{hypoxic}^\sigma$ and decay with rate λ_N^σ , in the necrotic region. Under steady-state conditions, oxygen and nutrient uptake and decay is ^{5, 10, 11}:

$$0 = \nabla \cdot (D_\sigma \nabla \sigma) + \lambda_{ev}^\sigma(\mathbf{x}, t, \mathbf{1}_{vessel}, p_i, \sigma, h) - \lambda^\sigma(\sigma) \sigma \quad (4)$$

where $\mathbf{x}$ is position, t is time, $\mathbf{1}_{vessel}$ is the characteristic function for the vasculature (equals 1 at vessel locations and 0 otherwise), p_i is interstitial pressure, and h is the hematocrit in the vascular network ⁵. Extravasation is modulated by the extravascular interstitial pressure p_i , scaled by the effective pressure p_e with the weight of the convective transport component of small molecules, k_{p_i} ¹⁰:

$$\lambda_{ev}^\sigma = \bar{\lambda}_{ev}^\sigma \mathbf{1}_{vessel}(\mathbf{x}, t) \left(\frac{h}{H_D} - \bar{h}_{min} \right)^+ \left(1 - k_{p_i} \frac{p_i}{p_e} \right) (1 - \sigma) \quad (5)$$

$\overline{H}_D$ and h_{min} are constants representing normal and minimum hematocrit necessary for oxygen extravasation, respectively. $\overline{\lambda}_{ev}^\sigma$ represents the constant transfer rate from pre-existing and tumor-induced vessels.

The loss term represents the oxygen and nutrient uptake in the various tissue regions ⁵:

$$\lambda^\sigma = \begin{cases} \lambda_{host}^\sigma & \text{in host tissue} \\ \lambda_{prolif}^\sigma & \text{in proliferating tissue} \\ \lambda_{hypoxc}^\sigma & \text{in hypoxic tissue} \\ \lambda_{necrotic}^\sigma & \text{in necrotic tissue} \end{cases} \quad (6)$$

1.4 Tumor Tissue Proliferation

Assuming that tumor tissue proliferation is proportional to the local concentration of oxygen and nutrients, the non-dimensionalized net proliferation rate is as follows ¹²:

$$\lambda_p = \begin{cases} 0 & \text{in host tissue} \\ \sigma - A & \text{in proliferating tissue} \\ 0 & \text{in hypoxic tissue} \\ -G_N & \text{in necrotic tissue} \end{cases} \quad (7)$$

where A is the native apoptosis rate of tumor tissue and G_N is the rate of volume loss due to necrosis.

1.5 Effect of Chemotherapy

Cisplatin is a cell cycle dependent drug; hence, its cytotoxic effect is only exerted on proliferating cells. Accordingly, the drug effect is included into the proliferation term for λ_p ,

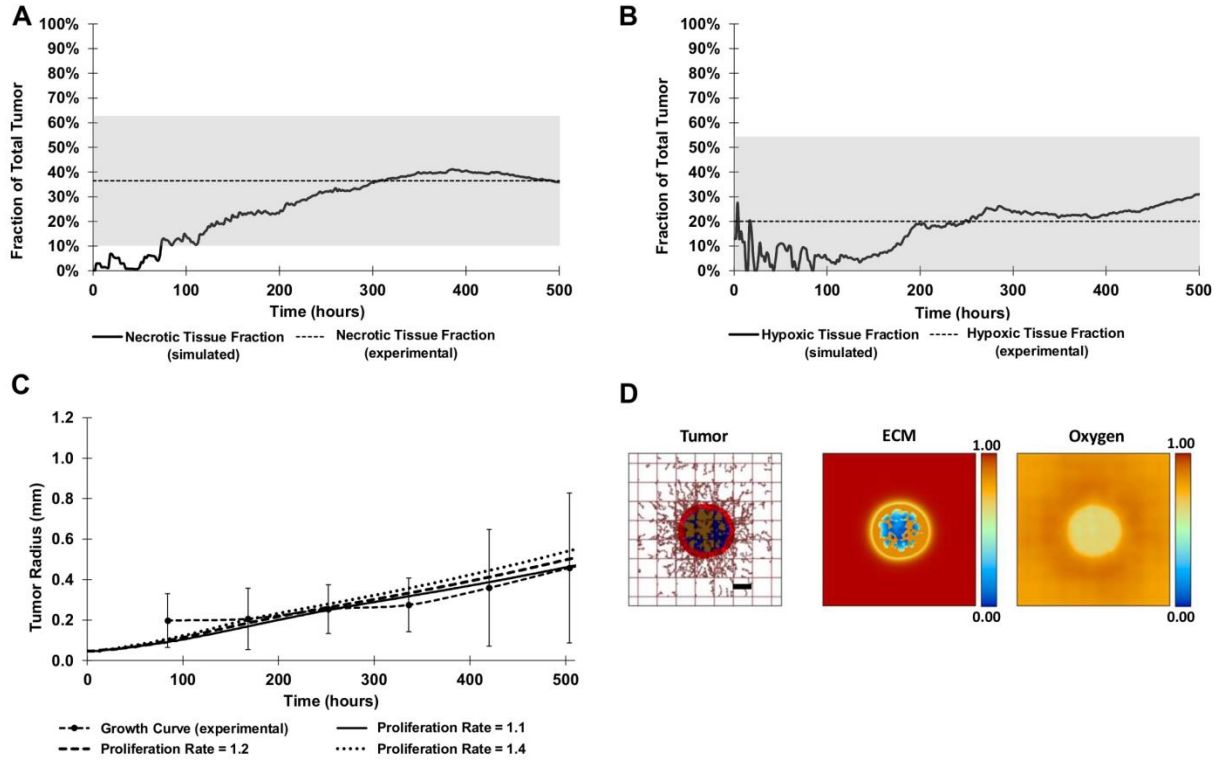
where $\overline{\lambda}_{effect}$ is the rate of drug-induced cell death ¹¹:

$$\lambda_p = \sigma(1 - \overline{\lambda}_{effect}c) - A. \quad (8)$$

This pharmacodynamic model assumes that cell death is instantaneous and that proliferation and apoptosis rates remain similar before and after the therapy.

2. Simulation of Baseline Tumor

To simulate a representative lung tumor nodule, the model parameters determining the proportion of hypoxic and necrotic tissue, as well as the growth rate were initially calibrated to lung tumor experimental data ^{9, 13, 15}. An average hypoxic fraction of 20% was found by Zegers et al. in 15 non-small lung cancer (NSCLC) patients via Positron Emission Tomography and Computed Tomography (PET-CT) imaging ¹⁵. In Pan et al., an average necrotic fraction of 36.5% was found in 14 patients with malignant lung masses ⁹. Growth rate was determined from Xu et al., in which a NSCLC xenograft model was measured in volume over a period of seven weeks ¹³. To reflect the values obtained from the literature, the model parameters were interpolated until the desired tumor characteristics were achieved, fitting the model to the experimental data, as shown in **Supplementary Fig 1**.



Supplementary Fig 1. Fitting of Model Growth to Experimental Data. (A) Necrotic tissue fraction fitted to experimental data ⁹. (B) Hypoxic tissue fraction fitted to experimental data ¹⁵. Gray areas represent the standard deviation of the experimental data. (C) Baseline tumor growth curve fitted to growth rate observed experimentally ¹³. (D) Simulation of tumor and vasculature, extracellular matrix (ECM) and oxygen of a representative NSCLC tumor at 400 h, grown using parameter values as in **Table 1**. In the tumor tissue, colors represent different regions of tumor tissue (red = proliferating, blue = hypoxia, brown = necrosis). Lines represent vasculature, with regular grid simulating pre-existing capillary network and irregular lines simulating neo-angiogenic vessels stimulated to grow towards tumor up the gradient of tumor angiogenic factors (TAF). ECM and oxygen color scales represent range from 0 to 1. Bar = 250 μ m.

The main model parameters and their baseline values are in **Supplementary Table 1**. The proliferation rate was calibrated as described in **Supplementary Fig 1**. Oxygen threshold for hypoxia denotes oxygen value at which tumor tissue ceases proliferation and becomes quiescent (hypoxic). Oxygen threshold for necrosis is oxygen value at which tumor tissue ceases being hypoxic and becomes necrotic. Oxygen diffusivity is used to non-dimensionalize the model parameters as noted in the Table. Oxygen transfer rate from vasculature and the various oxygen uptake rates were calibrated to achieve sustained tumor proliferation to a

distance of 100 μm from vasculature, matching typical distance of oxygen diffusion. Cisplatin diffusivity was estimated based on molecule size compared to oxygen, while decay rate was set based on the drug half-life ⁴. Tumor aggressiveness was set to achieve sustained tumor growth, matching previous calibration ¹². Lactic acid parameters were estimated based on their effect on tumor aggressiveness and their coupling to the macrophage polarization to Type 2. Lastly, macrophage polarization constants were set based on previous measurements ⁶.

Supplementary Table 1. Tumor Model Main Parameters. All other parameters are as in ¹².

Model Parameter	Parameter Value	Source
Tumor proliferation rate	1.1**	Calibrated to ¹³
Oxygen threshold for hypoxia	0.6540**	Calibrated to ¹⁵
Oxygen threshold for necrosis	0.6500**	Calibrated to ⁹
Oxygen diffusivity	1*	¹²
Oxygen transfer rate from vasculature	5*	¹²
Oxygen uptake rate by proliferating cells	1.5*	¹²
Oxygen uptake rate by hypoxic cells	1.3*	¹²
Oxygen uptake rate by tumor microenvironment	0.12*	¹²
Oxygen decay rate	0.35*	¹²
Cisplatin diffusivity (D_G)	0.6*	Estimated
Cisplatin decay rate (λ_{decay}^{-G})	0.5hr half-life	⁴
Tumor aggressiveness	60**	¹²
Lactic acid diffusivity	1	Estimated
Lactic acid uptake rate	0.12*	Estimated
Lactic acid washout by vasculature	0.1*	Estimated
Type 1 macrophage polarization constant	20**	⁶
Type 2 macrophage polarization constant	12**	⁶
* Value is rescaled by the square of the simulation system characteristic length (1cm) and divided by the system characteristic time (1s) multiplied by the oxygen diffusivity ($1 \times 10^{-5} \text{cm}^2 \text{s}^{-1}$) ¹² ** Used for baseline tumor parameter values		

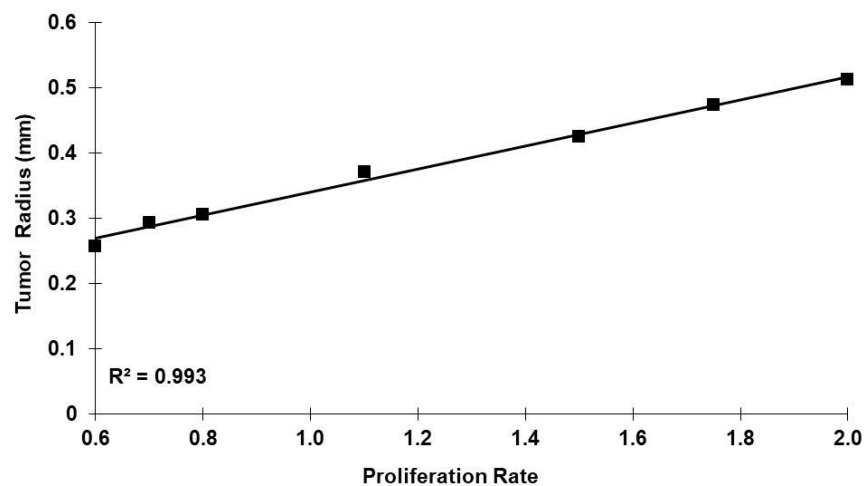
3. Determination of Range of Model Parameter Values

3.1 Variation in Tumor Proliferation Rate

The upper and lower bounds for the range of proliferation rate constant values were calibrated by interpolation of values that spanned biologically-relevant cancer cell proliferation rates.

Tumor radius at 400 h with various values of proliferation rate constant is shown in

Supplementary Fig 2. A minimum value of 0.6 for the proliferation rate constant was chosen as the lower bound, and a maximum value of 2 was chosen as the upper bound. The increase in tumor radius over time remains relatively linear among all cases, as expected.



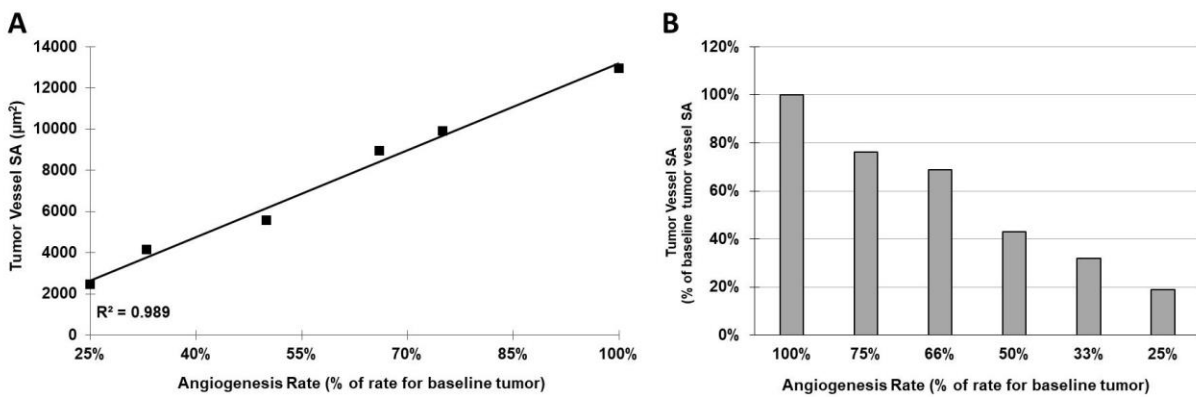
Supplementary Fig 2. Determination of range for tumor proliferation rate. Tumor radius vs. proliferation rate (at 400 h). A value of 0.6 was chosen as the lower bound and 2.0 as the upper bound.

3.2 Variation in Tumor Vascularization

In order to account for metabolites known to affect tumor angiogenesis, a parameter was implemented within the angiogenesis model to allow for a decrease in the tumor vessel density below the baseline value, set in the original model to match a highly vascularized tumor, such as glioblastoma¹². The parameter allows control over the rate of angiogenesis without affecting the TAF production rate or ECM degradation. There are alternative ways to control angiogenesis, but this approach was chosen for simplicity to yield the desired control. After the

probability of sprout tip movement in one of the 4 directions based on local TAF concentration is determined ⁵, the parameter decreases the frequency in which the vessel growth will activate in the determined direction. This decrease does not affect the probability of vessel growth, mitosis, or the semi-stochastic nature of the model. In **Supplementary Fig 3A**, the parameter interpolation is shown as tumor vessel surface area vs. angiogenesis rate at 400 h.

Supplementary Fig 3B shows that decreasing angiogenesis rates (with respect to angiogenesis rate for baseline tumor) corresponded to decreasing tumor vessel SA.



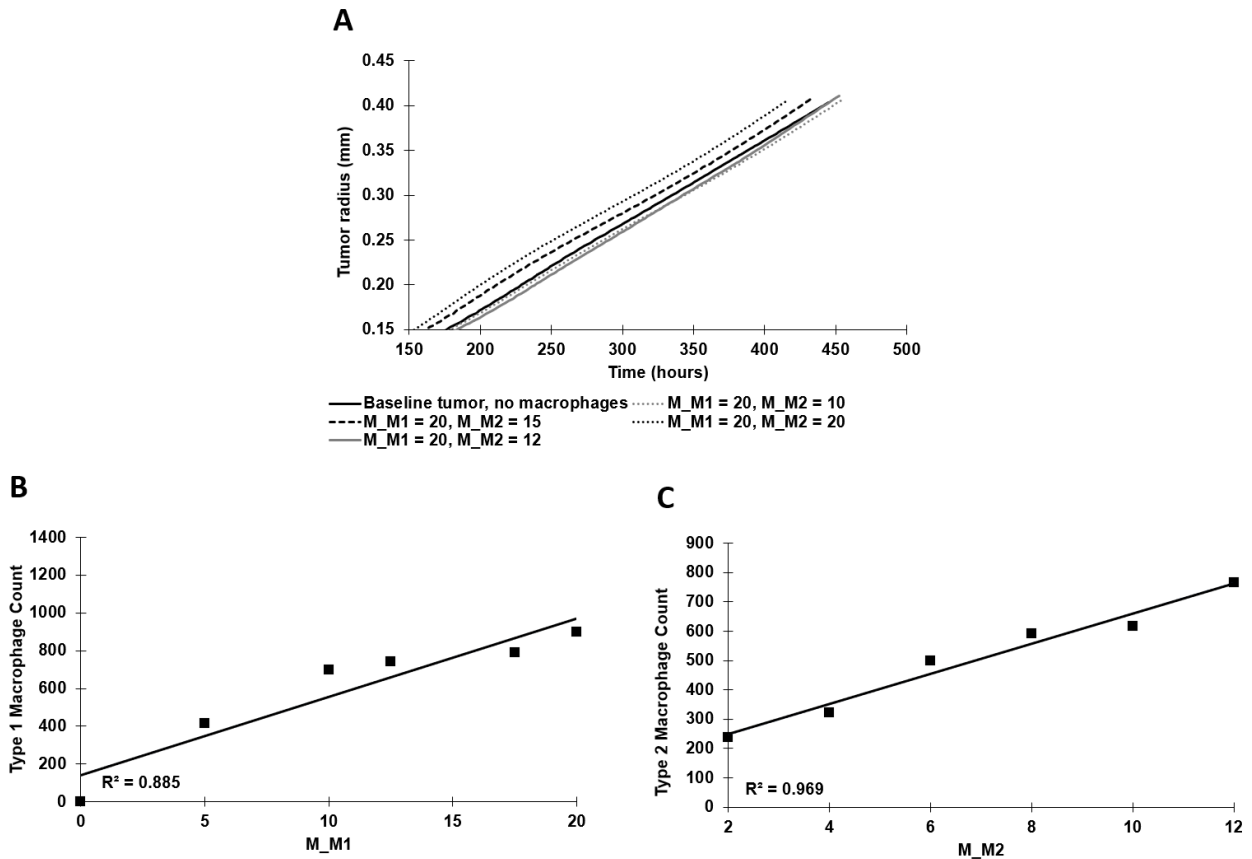
Supplementary Fig 3. Variation in Tumor Vascularization. (A) Parameter interpolation curves for the rate of angiogenesis compared to the baseline case, showing (A) tumor vessel surface area (SA) vs. angiogenesis rate (at 400 h) and (B) tumor vessel SA at a tumor radius of 0.4mm.

3.3 Variation in Macrophage Polarization

Immune system activity is simulated by the Type 1 (M1) and Type 2 (M2) macrophages in the model. Constants M_{M1} and M_{M2} affecting polarization to an M1 or M2 phenotype were calibrated using the baseline tumor parameters. As shown in ⁶, inclusion of macrophages dynamically affects the hypoxic and necrotic tumor tissue fractions. The baseline tumor growth (**Supplementary Fig 1**) was matched with values of $M_{M1} = 20$ and $M_{M2} = 12$

(**Supplementary Fig 4A**), yielding a tumor radius matching the radius within 1.6% of the baseline case at 400 h. The upper bound for maximum pro-tumorigenic activity was found by deactivating all Type 1 macrophages ($M_{M1} = 0$) and leaving all Type 2 macrophages at the

baseline probability constant ($M_M2 = 12$) (**Supplementary Fig 4B**). In **Supplementary Fig 4C**, a minimum value of $M_M2 = 2$ was chosen as the lower bound for which tumor growth occurred with minimal immune pro-tumorigenic input.



Supplementary Fig 4. Calibration of macrophage polarization to maintain tumor growth consistent with baseline case and interpolation of upper and lower bounds. (A) Variation in probability of conversion to Type 2 (M_M2). Value of Type 2 macrophage polarization constant $M_M2 = 12$ matched the radius to within 1.6% of the baseline tumor at 400 h while holding Type 1 macrophage polarization constant M_M1 constant at a value of 20. (B) Interpolation of upper and lower bounds of M_M1 at 400 h. Value of $M_M1 = 0$ was chosen as the lower bound and $M_M1 = 20$ was chosen as the upper bound. (C) Interpolation of upper and lower bounds of M_M2 at 400 h while holding M_M1 constant at a value of 20. Value of $M_M2 = 2$ was chosen as the lower bound and $M_M2 = 12$ as the upper bound.

3.4 Effect of Metabolites on Tumor Vascularization

The effect of metabolites on tumor vascularization is taken into account by following the proposed method to calibrate the rate of angiogenesis for any particular sample based on the

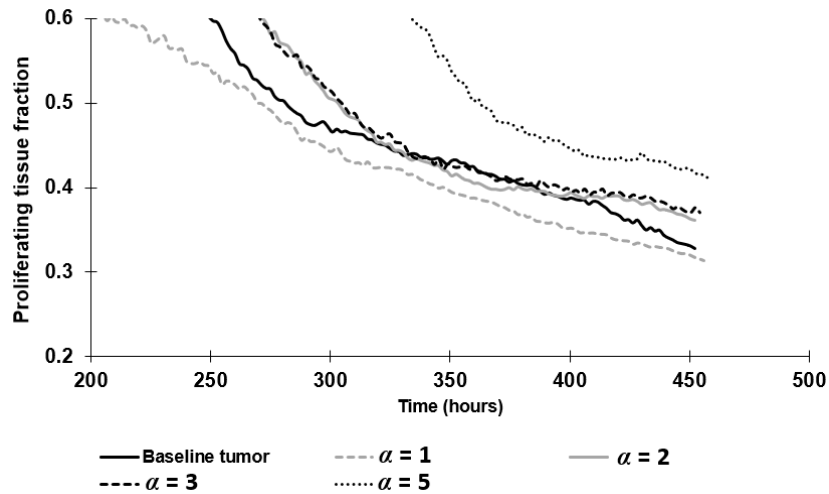
metabolites known to affect tumor angiogenesis^{1-3, 7, 14}, including acetyl-L-carnitine, fumarate, lactic acid, proline and pyruvate (**Table 1**).

3.5 Coupling of Lactic Acid to Tumor Aggressiveness

The tumor aggressiveness parameter G was coupled to the concentration of lactic acid within the tumor boundary. This allows the concentration of lactic acid to modulate the tumor growth:

$$G = \left(\frac{G_{baseline}}{L_{midpoint}} \right) L_{tumor} + a \quad (9)$$

where G is tumor aggressiveness, L_{tumor} is the total lactic acid within the tumor boundary at a given time, and a is a summation constant. The value of the coefficient multiplying the total lactic acid was found by dividing the baseline tumor aggressiveness value ($G_{baseline}$, 60) by the concentration of lactic acid at the mid-point of the growth phase of the baseline case ($L_{midpoint}$, non-dimensionalized value of 4,400). The value of the summation constant a was interpolated to find the proliferating tissue fraction that aligned (within 1.5%) with the baseline case at 400 h (**Supplementary Fig 5**). A summation constant of $\alpha=2$ was determined to match most closely to the baseline tumor.



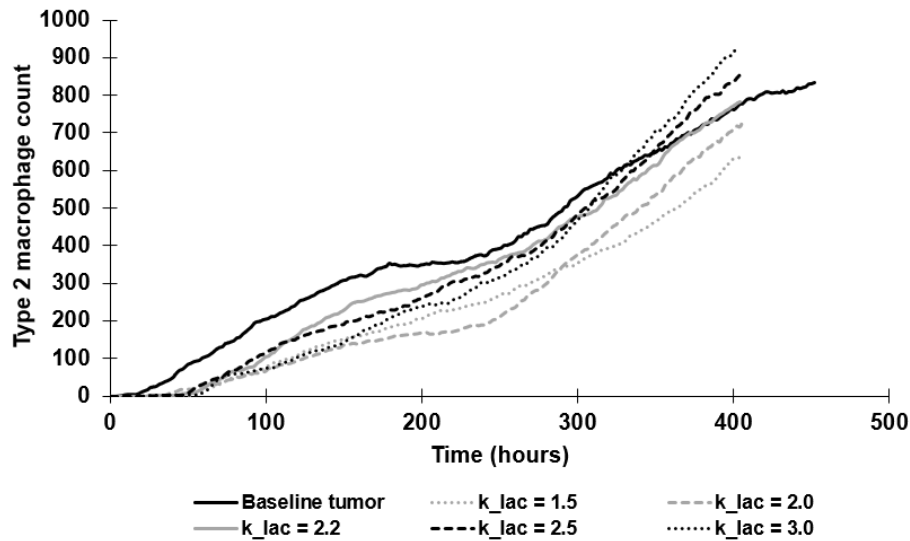
Supplementary Fig 5. Calibration of the summation term in coupled lactic acid and tumor aggressiveness equation. After including the equation in the model, a summation term (α) of +2 resulted in a proliferating tissue fraction within 1.5% of the baseline tumor at 400 h with an aggressiveness value of 60.

3.6 Coupling of Lactic Acid to Macrophage Polarization

Type 2 macrophage polarization was coupled to lactic acid concentration by multiplying the coefficient k_{lac} by local lactic acid concentration L as follows:

$$R_{M2} \propto k_{M2} \cdot k_{lac} \cdot L \cdot (C_{M2f} + k_{T2M2} \cdot C_{IL-10}) \quad (10)$$

The coefficient k_{lac} was calibrated by finding a value for which the Type 2 macrophage count matched that of the baseline tumor (**Supplementary Fig 6**). At 400 h, the total number of Type 2 macrophages in the case $k_{lac} = 2.2$ was within 2.4% of the count for the baseline case.

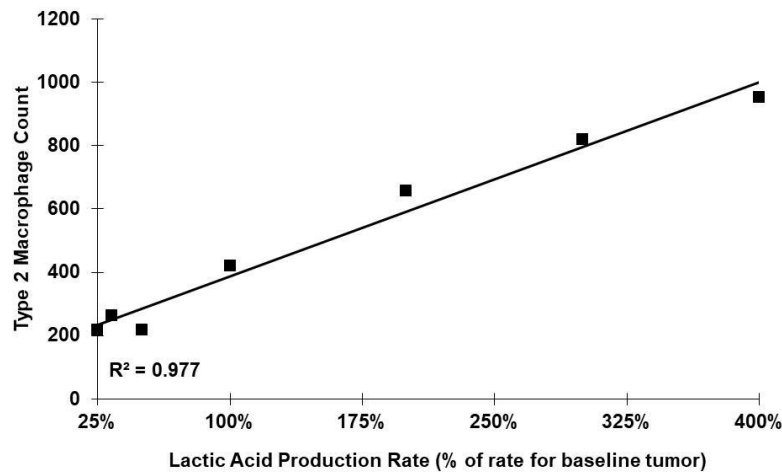


Supplementary Fig 6. Interpolation curves for the value of the coefficient coupling lactic acid with Type 2 macrophage polarization. Comparison to baseline, showing total Type 2 macrophage count over time. A coefficient value of $k_{lac} = 2.2$ was within 2.4% of the count for the baseline case at 400 h.

3.7 Variation in Lactic Acid Production

Once the tumor aggressiveness and Type 2 macrophage polarization were coupled to the local concentration of lactic acid, the production rate was varied to be able to represent heterogeneity in tumor metabolism. To prevent over-fitting of the model, the range of lactic acid production rate was chosen to be 25% to 400% of the baseline case (1/4x to 4x). The Type 2 macrophage

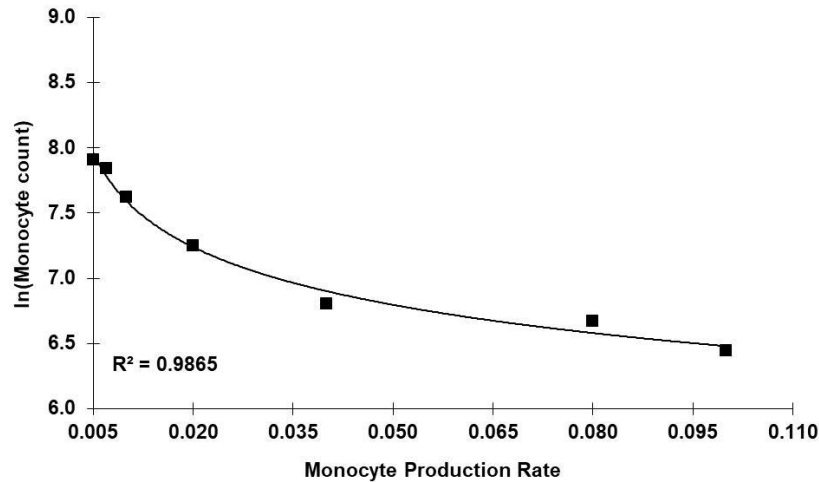
population at 400 h within the host and tumor tissue increased proportionally as the production rate rose in time, as expected (**Supplementary Fig 7**).



Supplementary Fig 7. Calibration of lactic acid production rate. Effect of lactic acid production rate variation on the number of Type 2 macrophages within the tumor boundary (at 400 h).

3.8 Variation in Monocyte Production

The production rate (rate of appearance) of monocytes was varied to represent heterogeneity in immune cell activation. The range of monocyte production was chosen to be 50% to 200% of the baseline tumor value (1/2x to 2x). The monocyte population at 400 h within the host and tumor tissue decreased exponentially as the production rate constant increased (**Supplementary Fig 8**).

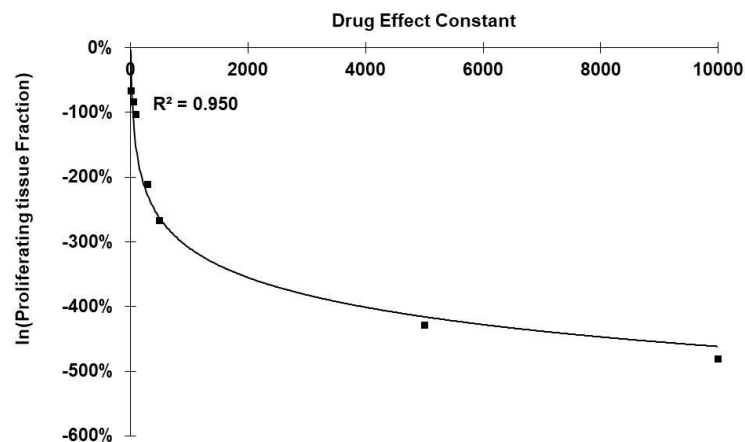


Supplementary Fig 8. Determination of range for monocyte production rate. Natural log of monocyte population vs. monocyte production rate constant (at 400 h). A value of 0.005 was chosen as the lower bound and 0.100 as the upper bound.

3.9 Variation in Drug Effect

The drug effect was varied to represent heterogeneity in tumor drug resistance. Cisplatin was simulated with a half-life of 30 minutes. The proliferating tissue fraction immediately following a bolus drug injection decreased exponentially as the drug effect constant increased

(Supplementary Fig 9).



Supplementary Fig 9. Determination of range for drug effect constant. Natural log of proliferating tissue fraction vs. drug effect constant (at one discrete time point after systemic drug injection). A value of 10 was chosen as the lower bound and 10,000 as the upper bound.

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